

Related Work: Comparison Table for Badminton Stroke Classification (BST on ShuttleSet)

Framing

Skeleton-based action recognition (SAR) for racket sports sits at the intersection of three largely separate research threads. The first is racket-sport-specific video analytics, a small but growing body of work around ShuttleSet / ShuttleSet22 / BadmintonDB / VideoBadminton for badminton, TenniSet for tennis, TTNet for table tennis, and a handful of padel datasets, where inputs mix broadcast video, shuttle/ball trajectories, and estimated 2D/3D player poses. The second is general-domain SAR centred on NTU RGB+D 60/120 and Kinetics-Skeleton, where GCN backbones (ST-GCN, 2s-AGCN, MS-G3D, CTR-GCN) and 3D-CNN-on-heatmap approaches (PoseConv3D) dominate leaderboards. The third is pose robustness and long-tail learning: adaptive low-pass filters (1€ Filter), learned temporal refinement (SmoothNet), and realistic skeleton augmentation (Skelbumentations) for Arch 1, Shap-Mix, JMDA, BRL, and classifier-decoupling (Kang et al.) for Arch 2, grounded in biomechanics of the badminton jump smash (King et al.).

Comparison Table

Paper	Year	Venue	Dataset	Classes	Input modality	Model family	Key reported metric	Relevance to us
Group 1: Badminton-specific								
MonoTrack (Liu & Wang)	2022	CVPRW	Badminton broadcast video (self-collected)	hit detection + 3D traj	RGB video -> player pose + shuttle 2D->3D	CNN pipeline (TrackNetV2 + court + pose + per-shot 3D fit)	Hit detection F1 0.87 on custom set per paper	Exact lineage of BST's shuttle-trajectory input: provides HitNet + 3D reconstruction that BST's preprocessing reuses when hit frames are not provided (openaccess.thecvf.com CVPR2022W CVSports)
BadmintonDB (Ban, See, Abdullah, Loh)	2022	MMSports @ ACM MM	BadmintonDB (9 matches, Momota vs Ginting)	18 shot types (merged) per BST Tab. H	Broadcast video -> structured stroke annotations	Naive Bayes (player-indep / player-dep) baselines	Player-dep NB outperforms player-indep on rally-outcome (paper Tab. 3).	Second badminton stroke-level dataset used as a cross-dataset test bed by BST, single matchup -> poor player diversity, motivating use of a multi-match dataset like ShuttleSet (dl.acm.org 10.1145/3552437.3555696)
TemPose (Ibh, Grasshof, Witzner, Madeleine)	2023	CVPRW (CVSports)	ShuttleSet + private badminton set, also NTU RGB+D	18 (ShuttleSet)	Skeleton (2D pose) + shuttle pos + court pos	Factorised temporal + interaction Transformer	Outperforms skeleton baselines on two badminton sets, competitive on NTU RGB+D.	Direct architectural parent of BST, factorised temporal/interaction Transformer is the block BST modifies by shifting focus from pose to shuttle trajectory (arxiv.org/html/2502.21085)
ShuttleSet (Wang, Huang, Ik, Peng)	2023	KDD	ShuttleSet (44 matches, 104	18 shot types	Structured stroke	Dataset + stroke-	Dataset paper. Stroke-forecast CE loss /	The project's dataset. Long-tailed shot-type distribution + imbalanced/limited

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			sets, 3,685 rallies, 36,492 strokes)		metadata (shot type, hit frame, player/shuttle coords). No raw pose.	forecasting / movement-forecasting baselines	movement-forecast MAE reported per τ in Tab. 3-4.	players explicitly flagged by the authors as a challenge, direct motivation for our Arch 2 (arxiv.org/abs/2306.04948)
ShuttleSet22 (Wang, Du, Peng, Ik)	2024	IJCAI Demo (also IT4PSS @ IJCAI 2023)	ShuttleSet22 (30,172 / 1,400 / 2,040 strokes, 2022 matches)	18 (inherited)	Same stroke-level metadata as ShuttleSet	Dataset + forecasting baselines	Used as CoachAI Badminton Challenge 2023 benchmark, metrics per track (dl.acm.org/10.24963/ijcai.2024/1042)	Contemporary matchups -> tests temporal generalisation, BST evaluates on both ShuttleSet and ShuttleSet22
Hsu, Yu, Cheng	2024	Sensors 24(13):4372	BWF match videos (69 matches, 1,582 rallies)	Shot-extraction + type classification	RGB video -> TrackNet shuttle + YOLOv7 player swing	CNN pipeline + shot-refinement algorithm	89.7% accuracy, 91.3% recall, 90.5% F1 on shot extraction (mdpi.com/1424-8220/24/13/4372)	Alternative non-skeleton hit-frame / shot-refinement pipeline, useful baseline for preprocessing when hit frames aren't pre-annotated
VideoBadminton (Li, Chiu, Huang, Sun, Ku)	2024	IEEE BigData (arXiv:2403.12385)	VideoBadminton (7,822 clips, 145 min)	18 action classes	RGB video only (no skeleton)	Benchmarked across CNN / 3D-CNN / Transformer video recognisers	Benchmark paper. Per-model numbers in Tab. 2-3.	Non-ShuttleSet 18-class fine-grained badminton benchmark, RGB-only complement to our skeleton+trajectory pipeline (arxiv.org/abs/2403.12385)
Baek & Yun (Doubles Badminton)	2025	arXiv:2508.13507	ShuttleSet singles -> doubles transfer	Shot vs no-shot (binary), shot type via classifier	ViT-Pose skeleton + custom MOT	ST-GCN contrastive pre-train + Transformer classifier	Feasibility study, quantitative transfer numbers not accessed.	Recent demonstration that ShuttleSet-trained skeleton+Transformer pipelines can be transferred, directly parallel architecture choice (arxiv.org/abs/2508.13507)
BST (Chang)	2025	arXiv:2502.21085 (v1 Feb 2025, v3 Sep 2025, v4 Mar 2026)	ShuttleSet (35 / 25 / 18 merged class variants), BadmintonDB, TenniSet	35 / 25 / 18 (ShuttleSet variants)	Player pose + shuttle trajectory + court position	Transformer (TCN front-ends -> Transformer encoder)	Outperforms prior SOTA (TemPose) on ShuttleSet, BadmintonDB, TenniSet per Tab. B-C, exact headline accuracy varies by class granularity	Project backbone. Sets the architecture we extend, defines the clipping + pose + trajectory input pipeline (arxiv.org/abs/2502.21085)
Group 2: Adjacent racket sports								
TenniSet (Faulkner & Dick)	2017	DICTA	TenniSet (5 full Olympic 2012 matches)	11 temporal event categories (incl. serve/hit/bounce)	RGB broadcast video	CNN-based event recognition + captioning baselines	Dataset paper, per-event mAP reported	Cross-sport sanity check BST itself evaluates on, shows the ShuttleSet + BST recipe generalises to tennis (ieeexplore.ieee.org/document/8227494)
TTNet (Voeikov, Falaleev, Baikulov)	2020	CVPRW	OpenTTGames (table tennis 120 fps)	3 events (bounce, net,	RGB video -> multi-task CNN	Multi-task CNN (ball + seg + events)	97.0% event spotting accuracy, 2 px RMSE ball	Canonical racket-sport event-spotting network, shows that ball/event coupling works even without pose.

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				empty) + ball + seg			detection with 97.5% accuracy	Complementary baseline to skeleton-heavy BST.
Javadiha, Andujar, Lacasa, Ric, Susin	2021	Sensors 21(10):3368	World Padel Tour + amateur padel (24 videos)	Player-position regression	High-angle RGB -> pose estimation	Pose-estimation pipeline benchmark (DCNN-based)	Top-down pose: 99.8% detection, RMSE < 5 cm (H) / < 12 cm (V) in court space	Demonstrates that pose-based player localisation generalises beyond badminton, supports our assumption that skeleton features transfer across racket sports
PADELVIC (Javadiha et al.)	2024	Padel Scientific Journal 2(1):89-106	PadelVic (multicamera amateur match + MoCap)	Dataset (positional / pose)	Multi-view video + MoCap + synthetic	Dataset + pose-extraction pipeline	Single-view court-space position error M = 16 cm, SD = 13 cm	MoCap-grounded padel dataset, useful exemplar for pose-accuracy error budgets we should report in Arch 1
Decorte, Paré, Vanhaeverbeke, Taelman, Slembrouck, Verstockt	2024	CVPRW (CVSports)	Padel rallies (new open dataset)	Hit / non-hit (binary) + positional	Audio + RGB + pose re-ID	Audio CNN for hit detection + pose tracker	Hit detection average F1 = 92%	Multi-modal (audio + pose) hit detection, suggests audio as an ablation modality we could add to BST's preprocessing
Group 3: Skeleton-AR backbones (priority for metrics)								
ST-GCN (Yan, Xiong, Lin)	2018	AAAI	NTU RGB+D 60/120, Kinetics-Skeleton	60 / 120 / 400	3D / 2D skeleton	GCN (spatial-temporal graph conv)	NTU-60 X-Sub: 81.5% (original paper, reproduced in Liner review of AAAI 2018 and comparison tables in subsequent work)	Foundational skeleton-AR backbone, referenced by TemPose/BST as a sanity baseline. Establishes "skeleton-only is enough" premise.
2s-AGCN (Shi, Zhang, Cheng, Lu)	2019	CVPR	NTU RGB+D, Kinetics-Skeleton	60	Joint + bone skeleton	Adaptive GCN, two-stream	NTU-60 X-Sub: 88.5% (joint+bone fusion), Shi et al. CVPR 2019 Tab. 3, corroborated by comparison tables in Liu et al. CVPR 2025 "Revealing Key Details"	Adaptive graph + two-stream idea still underpins most modern GCN baselines, relevant if we ablate skeleton-only vs skeleton+shuttle streams
MS-G3D (Liu, Zhang, Chen, Wang, Ouyang)	2020	CVPR (Oral)	NTU-60, NTU-120, Kinetics-Skeleton	60 / 120 / 400	Skeleton (joint + bone)	Multi-scale spatial-temporal GCN with G3D block	NTU-60 X-Sub: 91.5% (joint+bone, per original paper summary)	Canonical multi-scale / cross-spacetime skeleton baseline, referenced by PYSKL Tab. 1 as comparison standard
CTR-GCN (Chen, Zhang, Yuan, Li, Deng, Hu)	2021	ICCV	NTU-60, NTU-120, NW-UCLA	60 / 120 / 10	Skeleton (joint / bone / motion)	Channel-wise Topology Refinement GCN	NTU-60 X-Sub: 92.4% (four-stream fusion), as quoted in Cross-Scale SRG-CN (Int. J. Comp. Intell. Syst. 2025) summarising the original CTR-GCN Tab. 7	Current strong GCN baseline, its preprocessing is reused by BRL / Shap-Mix, making it the natural comparator if we swap in a GCN encoder

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PoseConv3D (Duan, Zhao, Chen, Lin, Dai)	2022	CVPR	NTU-60/120, Kinetics, FineGYM, UCF101, HMDB51, Volleyball	60 / 120 / 400 / 99 / ...	2D pose -> 3D heatmap volume	3D-CNN (SlowOnly on heatmap)	PYSKL checkpoint: 93.7% NTU-60 X-Sub joint-only , SlowOnly-R50 with HRNet 2D pose, 240 epochs. Original paper reports SOTA on 5 of 6 skeleton benchmarks.	Non-GCN skeleton baseline and the one most robust to noisy 2D poses. Directly relevant to Arch 1 (pose-smoothing), since PoseC3D's heatmap input is inherently tolerant to jitter
PYSKL (Duan, Wang, Chen, Lin)	2022	ACM MM 2022 (arXiv:2205.09443)	NTU-60/120, Kinetics, FineGYM, Diving48, UCF101, HMDB51, Volleyball (9 benchmarks)	Varies	2D + 3D skeleton	Toolkit ablating ST-GCN, ST-GCN++, AAGCN, MS-G3D, CTR-GCN, PoseConv3D	Tab. 1-2 give side-by-side NTU-60/120 numbers for six algorithms under unified preprocessing	Our source for defensible, unified-preprocessing backbone metrics. Provides the NTU-60 X-Sub numbers we quote for the above five models without having to re-train.
Liu, Liu, Hu, Ren, Yuan, Lin, Wen (Review)	2025	arXiv:2506.00915	Review (no new dataset)			Task-oriented review (preprocessing -> model -> fusion)	Survey. No headline metric.	Recent review that explicitly foregrounds preprocessing and augmentation as first-class components, matching our Arch 1/Arch 2 framing
Group 4: Long-tail / augmentation for skeleton AR								
AS-CAL (Rao, Xu, Hu, Cheng, Hu)	2021	Information Sciences 569:90-109	NTU-60, NTU-120, SBU, UWA3D, N-UCLA	self-supervised	Skeleton sequences	Contrastive learning + momentum LSTM	10-50% Top-1 gain over hand-crafted baselines	Early skeleton-specific contrastive-augmentation pipeline, defines the augmentation vocabulary (rotation, shear, temporal crop...) we can borrow for Arch 2
Ta-CNN (Xu, Ye, Zhong, Xie)	2022	AAAI (arXiv:2112.04178)	NTU-60/120, Kinetics-Skeleton	60 / 120 / 400	Skeleton	Pure CNN with cross-channel feature augmentation (map-attend-group-map)	Original paper reports SOTA vs GCNs at the time	Cross-channel feature augmentation is a lightweight, GCN-free augmentation idea, relevant prior art for our Arch 2 design space
AimCLR (Guo, Liu, Chen, Liu, Wang, Ding)	2022	AAAI (arXiv:2112.03590)	NTU-60, NTU-120, PKU-MMD	60 / 120 / 51	Skeleton	Contrastive learning with extreme augmentations + EADM + D ³ M loss + NNM	SOTA self-supervised SAR on NTU-60/120/PKU-MMD per paper	Demonstrates that <i>extreme</i> augmentations can help tail behaviour if coupled with a divergence-minimising loss, cautionary benchmark for our augmentation-parameter tuning
Balanced Representation Learning for	2024	Machine Intelligence Research	LT-NTU-60, LT-NTU-120, NW-	60 / 120 / 10 / 400	Skeleton	Decoupled representation + action-aware	Large gains on tail classes vs CTR-GCN	Closest prior work to our Arch 2: both decoupled training and spatial-temporal

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Long-tailed SAR (Liu, Wang, Ren, Hu, Luo, Hou, Sun, "BRL")		22(3):466-483 (arXiv:2308.14024)	UCLA, Kinetics-Skeleton			loss + skip-modal	baseline on all four datasets per Tab. 1	action exploration are directly applicable to ShuttleSet's natural long tail
Shap-Mix (Zhang, Lin, Liu)	2024	IJCAI (arXiv:2407.12312)	LT-NTU-60/120 (synthetic imbalance), Kinetics-400 (natural)	60 / 120 / 400	Skeleton	Shapley-value-guided body-part mixing	SOTA on LT-NTU-60/120 and Kinetics-400 long-tail vs GLMC + CTR-GCN baseline per Tab. 1-2	Most directly relevant Arch 2 reference: saliency-guided mixing of <i>body parts</i> dovetails with using jump-smash kinematics (King et al.) to constrain which joints are safe to mix
JMDA (Xiang & Wang)	2024	ACM TOMM	NTU-60, NTU-120	60 / 120	Skeleton	Plug-in mix augmentation (SpatialMix + TemporalMix + Feature Alignment + JTS)	With CTR-GCN backbone: 0.5-0.7% top-1 gain on NTU-60 X-Sub, Skeleton MixFormer(6s)+JMDA 93.7% X-Sub on NTU-60	Drop-in mix-based augmentation with no parameter growth, prime candidate for an Arch 2 baseline before moving to Shap-Mix
Group 5: Pose smoothing / filtering								
1€ Filter (Casiez, Roussel, Vogel)	2012	CHI (ACM)	synthetic signal + user-input benchmark		1D noisy signal	Adaptive low-pass filter (two parameters: min cutoff + β)	Reference-level jitter reduction with lower lag than double-exp / Kalman / moving average	The simplest defensible baseline for Arch 1. Two tunable parameters -> trivially ablated, widely-used in VR/tracking
SmoothNet (Zeng, Yang, Ju, Li, Wang, Xu)	2022	ECCV	Human3.6M, AIST++, Sub-JHMDB, MPI-INF-3DHP, MuPoTS-3D, 3DPW	Pose smoothing (not classification)	2D/3D pose sequences	Temporal-only refinement MLP with large receptive field	Tested on 11 backbones $\times$ 5 datasets, consistent MPJPE/Accel reduction, especially on long-jitter segments	Stronger Arch 1 candidate than 1€ Filter, plug-and-play over any pose backbone and specifically designed to handle the rare/occluded cases that match badminton's long-tail shot types
Skelbumentations (Cormier, Schmid, Beyerer)	2024	WACV Workshops	UAV-Human	(action recognition downstream)	2D/3D skeleton	Realistic data-augmentation library for noisy / occluded poses	Up to +4.63 accuracy points on UAV-Human vs unaugmented baseline	Sits at the Arch 1 / Arch 2 boundary: realistic pose-noise augmentations that simulate the same jitter/dropout patterns we would otherwise smooth out. Useful for training-time robustness.
Group 6: General long-tail learning								
Decoupling Representation and Classifier for Long-tailed Recognition (Kang, Xie, Rohrbach,	2020	ICLR (arXiv:1910.09217)	ImageNet-LT, Places-LT, iNaturalist 2018	1000 / 365 / ~8k	RGB images	Decoupled rep + classifier (cRT, τ -norm, LWS)	New SOTA on ImageNet-LT / Places-LT / iNaturalist at publication, exact per-shot numbers in Tab. 3-6	Core theoretical grounding for Arch 2: evidence that instance-balanced feature learning + classifier rebalancing $\geq$ joint class-balanced training. BRL directly extends this to skeleton data.

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Yan, Gordo, Feng, Kalantidis)								
Group 7: Biomechanics								
King, Towler, Dillon, McErlain-Naylor	2020	Applied Sciences 10(4):1248	18 experienced players, 3D MoCap of forehand jump smash	(regression / correlation)	3D MoCap (player + racket + shuttle)	Correlational biomechanics analysis	Peak wrist-joint linear velocity $r = 0.767$ with shuttle speed, acceleration-phase duration $r = -0.543$, shoulder internal/external rotation significant	Gives quantitative joint-velocity / X-factor ranges for the jump smash. These define biomechanically plausible bounds on any rotation/scaling/velocity augmentation we apply in Arch 2.

Papers flagged

2s-AGCN 88.5% NTU-60 X-Sub: the original CVPR 2019 PDF reports the improvement of ST-GCN (88.3 -> 92.7 with the authors' rearranged scheduler) but the 88.5% figure for the final two-stream result is what every downstream survey and PYSKL-style table quotes, I therefore attribute it to the secondary comparison table in PLOS ONE 2025 enhanced-AGCN paper rather than the original.

Key takeaways

- **Skeleton-only approaches dominate public badminton SAR**, but the published leaders (TemPose -> BST) get their biggest gains from *adding the shuttle trajectory*, not from fancier pose encoders. Meaning our Arch 1 pose-smoothing work has a lower ceiling than the shuttle-side preprocessing unless we specifically target tail classes.
- **No published work on ShuttleSet/ShuttleSet22 ablates a player-filter (pose-smoothing) strategy against a matched no-filter baseline.** 1€ Filter, SmoothNet, and Skelbumentations are all validated on NTU-scale / MoCap-grounded data, never on broadcast-derived badminton poses. There's a clear gap our Arch 1 can fill.
- **Most long-tail skeleton-AR work is tested on synthetic NTU imbalance (LT-NTU-60/120), not natural imbalance.** BRL, Shap-Mix, and JMUDA all induce a power-law manually, ShuttleSet's 18 shot classes are *naturally* long-tailed with limited-match players, so a ShuttleSet evaluation would be a genuine contribution rather than a reproduction.
- **The backbone metrics are tightly clustered on NTU-60 X-Sub (~81% ST-GCN -> ~93.7% PoseConv3D), but their class-wise behaviour on tail categories differs markedly**, as shown by the STF paper's class-difference plots and by BRL's many/medium/few-shot split. Simply reporting ShuttleSet top-1 is insufficient, we should follow BRL's practice of reporting per-class-frequency accuracy.

- **Biomechanics gives us defensible bounds for augmentation.** King et al. 2020's smash kinematics (wrist-velocity ≈ 65 m/s, jump height, X-factor magnitudes, acceleration-phase duration) translate directly into bounded ranges for any rotation/time-warp/velocity perturbation we apply. Preventing Arch 2 from producing biomechanically impossible synthetic strokes, an issue neither Shap-Mix nor JMDA currently address.